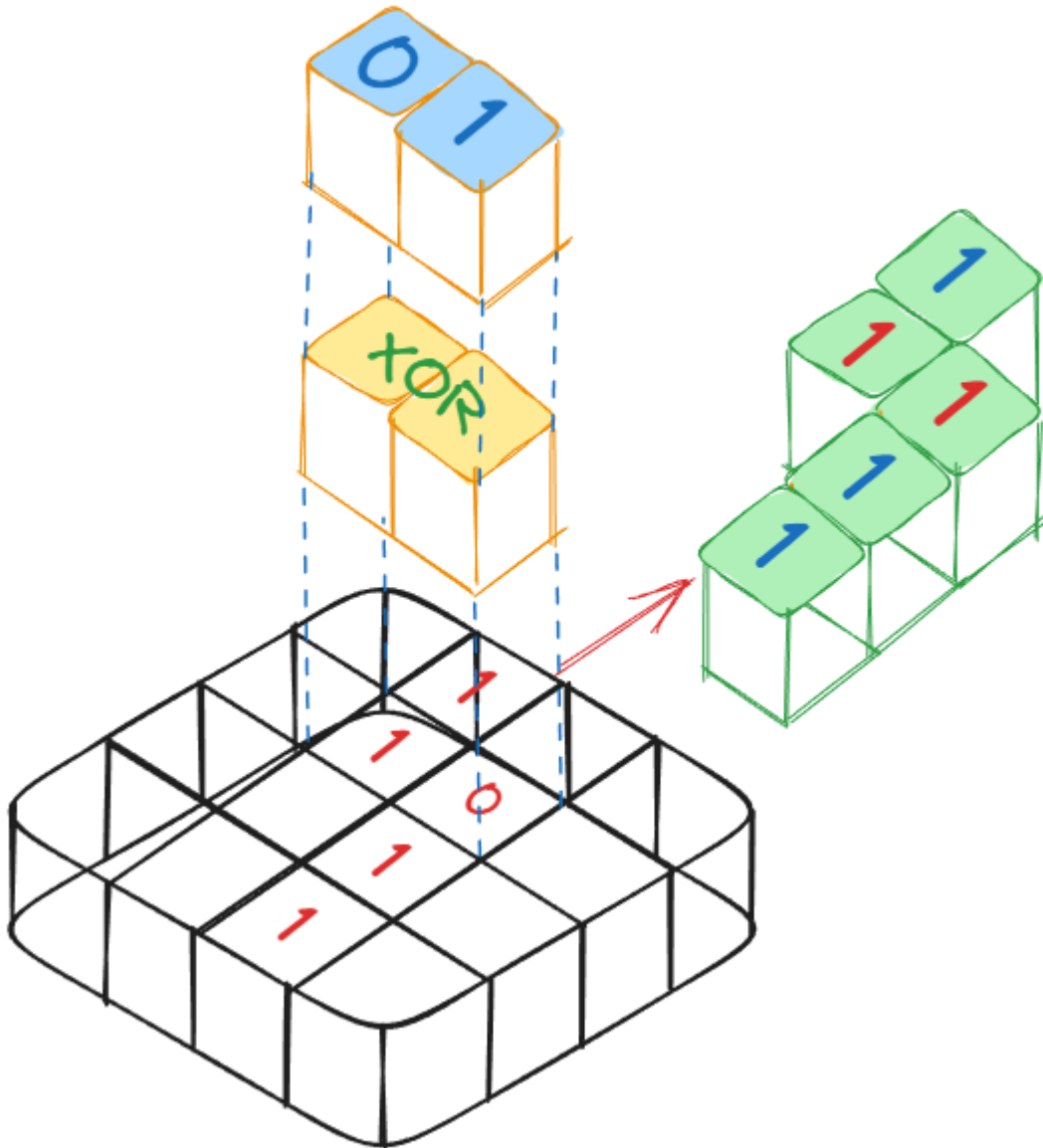


Leveraging VR Technologies for Edutainment in Learning Complex STEM Concepts Interim Report



COMP5925 Metaverse Project I Interim Report

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1.Introduction

In conventional classroom activities, abstract mathematical concepts – like discrete structures, path connectivity, and binary reasoning – are often poorly understood intuitively, with lower engagement. Given its spatial presence and embodied interaction, immersive virtual reality (VR) can be used to create a way to translate abstract symbolic rules into real-world, manipulable tasks. Inspired by this, we propose Binary Path, an entertainment-based VR puzzle game in which players lay “1-tiles” across a three-dimensional grid and create a single, connected path from the start to the goal. Through a quick “place-test-revise” circuit, the game builds path connectivity and binary logic into intuitive tactile, auditory, and visual experiences; while frequent feedback loops further solidify our grasp of the idea behind binary logic concepts. This project presents a gamified design approach in a way that promotes continuous experimentation and cumulative development of mathematical intuition through gamified learning processes. With this goal, BinaryPath should serve as a puzzle-based framework that allows adaptation to other VR educational scenarios. We expect that university students, beginners interested in binary logic and discrete structures, VR game players, and educators, researchers, developers, and game designers interested in how VR can model mathematical structures will get interest and understanding in this project.

2.Objectives

Objective 1 — Gamification Aligned with Learning Mechanics

We will achieve sustained engagement and concept formation by integrating task-embedded gamification that directly supports the *place-test-revise* loop: challenge calibration via tile budgets, meaningful penalties for illegal branches, micro-achievements for strategy discovery, and immediate multimodal feedback that affirms valid connectivity or flags misconceptions. Rather than adding extrinsic rewards on top of content, we embed motivational affordances into the structure of puzzle tasks so that progression reflects conceptual mastery, not mere time-on-task. This contributes a reusable design pattern for math-centric VR, advancing the mechanics-as-pedagogy paradigm and offering value to both educational VR and entertainment puzzle design. The approach builds on and extends gamification research showing improvements in interactivity, involvement, self-efficacy, engagement, focus, motivation, and academic outcomes when game elements are aligned with learning objectives (Nacke & Deterding, [2017](#); Bouchrika et al., [2019](#); Kim et al., [2017](#); Yildirim, [2017](#))

Objective 2 — Embodied Interaction with Immediate, Multimodal Feedback

We will harness VR’s core affordances—interaction, immersion, immediacy, involvement, and presence—to make abstract rules manipulable and perceptually vivid: in BinaryPath, learners directly place 1-tiles on a spatial grid and receive immediate validation via light-orb traversal, path highlighting, and subtle haptics, maintaining short loops and low failure cost to enable hypothesis testing under constraints and to transform symbolic connectivity/binary reasoning into embodied events that lower cognitive load (Lampropoulos et al., 2020; Freina & Ott, 2015; Psotka, 1995; Ryan, 2015; Sherman & Craig, 2003; Biocca & Delaney, 1995; Blascovich & Bailenson, 2011). This design aligns with evidence that safe, immersive, and interactive environments increase motivation, participation, and learning effectiveness and support experiential practice of complex skills across instructional settings (Bronack et al., 2006; Dickey, 2005; Dalgarno et al., 2011; Radianti et al., 2020; Annetta et al., 2009;). Our contribution and importance lie in presenting a pedagogy-aware interaction loop that couples embodied manipulation with instant state feedback—useful for micro-lessons and entertainment

puzzles alike—and in demonstrating a practical pathway for intuitive understanding of discrete structures in VR, where adoption continues to grow with improved accessibility (Hu-Au & Lee, 2017; Martín-Gutiérrez et al., 2017; Olmos et al., 2018; Merchant et al., 2014). Innovation over prior work stems from embedding immersion and immediacy into the rule-and-strategy formation itself—rather than layering extrinsic rewards—by fusing breadth-first search (BFS) rule-checking, world-space socket-based snapping, and multimodal cues into a coherent, high-responsiveness loop optimized for standalone headsets, thereby advancing mechanics-as-pedagogy for both VR in education and VR game design.

Objective 3 — A Modular, Quest-Ready VR Puzzle Framework

We will package Binary Path into a Unity/Quest-ready framework that cleanly separates grid generation, socket-based snapping, BFS connectivity checks, and multimodal feedback orchestration, exposing each as configurable modules and example scenes so educators, researchers, and indie teams can rapidly adapt the system to other math/logic topics (e.g., trees, flows, constraint satisfaction). Concretely, interaction will be implemented atop the XR Interaction Toolkit (Interactors/Intereactables and XR Socket Interactor for robust, world-space snapping), ensuring reproducible behaviours and clear affordances across devices. Cross-platform runtime will target OpenXR with Quest support to reduce vendor lock-in and align with current Unity guidance for Android/Quest deployment and controller profiles, while remaining compatible with Meta's feature set. To meet standalone HMD performance budgets, the framework will default to URP with SRP Batcher / GPU-Resident Drawer enabling GPU instancing via BatchRendererGroup to reduce draw calls and CPU time, with Forward+ path and documentation for when to enable/disable GRD in mobile builds; we will also provide Quest-focused presets and pointers to performance controls (e.g., target refresh rate, fixed foveated rendering, dynamic resolution) so adopters can tune for 72/80/90 Hz targets on device. At the system level, the framework contributes a reusable, mechanics-as-pedagogy template for short-cycle, puzzle-based learning consistent with design guidance from ScienceVR, which argues for explicit links between interaction design, in-app measurement, and learning outcomes in STEM VR; at the technology level, it consolidates widely scattered best practices (interaction, rendering, device configuration) into a coherent starter kit comparable in spirit to open-source XR scaffolds, but focused on *math-centric gameplay* and classroom deployment. By delivering a documented package (APIs, prefabs, validation tools) that is Quest-deployable out-of-the-box and OpenXR-aligned, this objective advances the state of practice for educationalVR (lowering replication cost, enabling rapid classroom pilots) and for VR games (codifying performant interaction/rendering patterns for puzzle loops on standalone headsets). (Planey, J., Trencovski, D., Kelle, S., 2026)

3.Literature Review

System perspective

Across STEM, immersive technologies tend to raise engagement and perceived understanding, yet consistent gains in learning effectiveness depend on task quality—namely manipulable representations, short feedback cycles, low cognitive load, and embedded assessment. Reviews also flag recurring shortcomings: many educational VR systems lack explicit links to learning theory and rely on usability-centric (not outcome-centric) evaluations, motivating designs that couple interaction, process data, and learning goals—the stance we adopt with BinaryPath's mechanics-as-pedagogy approach. In parallel, a large systematic review on VR + gamification shows that when challenge, progression, and feedback are intrinsically aligned with learning tasks (rather than added as extrinsic rewards), learners show higher motivation, engagement, self-efficacy, and achievement

across levels and subjects, which directly supports encoding connectivity and binary decisions inside a replayable puzzle loop.

Technology perspective

For robust interaction on standalone HMDs, Unity's XR Interaction Toolkit (Interactors/Interactables) and the XR Socket Interactor enable predictable world-space grabbing/snapping and clean event flows suited to a place-test-revise puzzle, while simplifying haptic/visual/audio orchestration. To sustain frame-time on Quest-class devices, a conservative rendering stack—URP with SRP Batchers and GPU-Resident Drawer (BatchRendererGroup-driven instancing) on Forward+—reduces CPU submission and draw calls (with documented trade-offs). We also expose refresh-rate and performance controls so projects can target 72/80/90 Hz and deploy dynamic resolution/FFR where appropriate; and we standardize on OpenXR (Quest Support) to minimize lock-in and streamline Android/Quest controller profiles. For rule checking, BFS provides $O(V+E)$ grid connectivity with pedagogically transparent frontier/visited visualization; and a recent three-level meta-analysis suggests that lightweight vibrotactile cues paired with clear state transitions can improve conceptual learning and UX in short sessions, reserving heavier kinesthetic solutions for future studies.

Conclusion

The literature converges on a practical mandate: encode the learning structure into the interaction structure. Yet reusable, math-centric VR templates remain scarce. BinaryPath responds with an entertainment-first, Quest-ready, OpenXR-aligned puzzle framework that operationalizes manipulability + rapid feedback + outcome-aware evaluation, addressing the system-design/tech-implementation/assessment coupling highlighted by prior frameworks and reviews.

4. Proposed Methodology

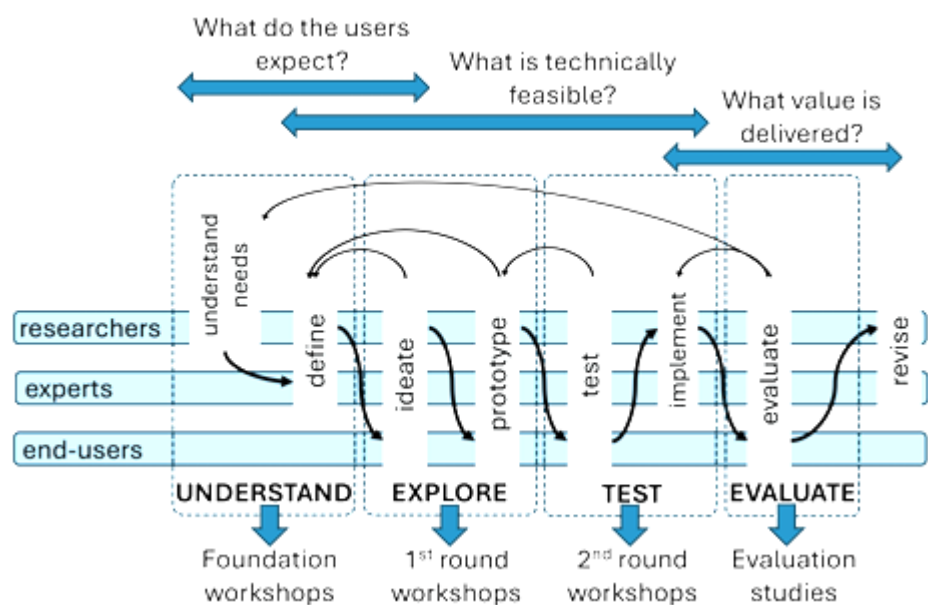


Figure 1. The process for designing the GLA (Papamitsiou et al., 2024).

The methodological approach of this project follows an iterative, human-centered design cycle inspired by the framework illustrated in Figure 1, which emphasizes continuous negotiation between user expectations, technical feasibility, and educational value throughout development. This process unfolds across four tightly connected phases—Understand, Explore, Test, and Evaluate—and involves three stakeholder groups: end-users, domain experts, and researchers. The following subsections describe how this cycle structures the design and implementation of BinaryPath.

Understand:

The initial stage focuses on identifying the conceptual and experiential needs related to path connectivity, binary reasoning, and discrete structure intuition. Drawing on paper mentioned above, we analyze how learners typically struggle with abstract symbolic reasoning, especially when feedback is delayed or non-interactive. This corresponds to the understand needs step in Figure 1, where researchers and end-users collaboratively articulate the learning barriers and expected forms of support.

The outcome of this phase is a set of design requirements, including:

- manipulable representations of connectivity,
- immediate multimodal feedback,
- low-friction interaction for novice VR users, and
- short, self-contained puzzle tasks suitable for micro-learning contexts.

Explore:

In the Explore phase, by iterates through cycles of ideation, sketching, and prototyping to translate the above requirements into concrete VR mechanics. Guided by the ideate-prototype loop in Figure 1, we experiment with different grid layouts, tile types, feedback modalities, and gesture/snap interactions using Unity's XR Interaction Toolkit.

Rapid prototypes are tested internally to refine:

- the place-test-revise loop,
- the clarity of connectivity rules,
- multimodal feedback (colour, haptic cues, audio), and
- early level structures that introduce constraints (tile budgets, branch penalties).

This stage serves to narrow the design space while ensuring technical feasibility on standalone HMDs (Quest-class devices).

Test:

Following early prototyping, the Test stage—aligned with prototype-test-revise in Figure 1—focuses on implementing a stable vertical slice of the BinaryPath system. This includes:

- the grid generator and tile placement logic,
- BFS-based connectivity validation,
- world-space UI and socket-based snapping,
- optimized URP rendering for stable 72–90 Hz performance, and
- short puzzle levels.

Small-scale playtests (1–3 participants per cycle) are conducted to observe discoverability, usability, and error patterns. Feedback drives continuous revision of interface clarity, difficulty tuning, and feedback timing. Each iteration aims to balance technical responsiveness with pedagogical transparency, ensuring that connectivity rules are both learnable and enjoyable.

Evaluate:

The final stage corresponds to the evaluate–revise loop in Figure 1 and centers on a structured pilot study to assess both the playability and initial learning value of BinaryPath. Evaluation focuses on:

- Usability and engagement: completion times, error profiles, subjective satisfaction, and perceived clarity of feedback;
- Player experience: flow, frustration, motivation, and sense of progression;
- System performance: frame-time stability, input latency, and comfort (zero-locomotion design).

Findings from this phase feed back into the refinement cycle, ensuring that the final system both supports enjoyable gameplay and serves as a reusable template for math-centric VR learning.

modelling:

By taking into account both time costs and technical capabilities, the project will utilize AI-generated methods to fulfill modeling requirements during the process. Through repeated comparison of the generated models’ effectiveness, the project will adopt the open-source Hunyuan2.1 model for model generation and application.

Table1. Comparison between 3DGS and diffusion model on the same image



AI using:

Generative AI tools will be applied in the analysis and maintenance phases of the project, and may also be used for code validation and review. However, no generative AI will be used for code writing in the project, nor for the preparation of related documentation. The models used in the project may be AI-generated, while all icons, music, sound effects, and text fonts will be sourced from content that is either copyright-free or already commercially licensed.

5. Concrete implementation

5.1 Core System Architecture

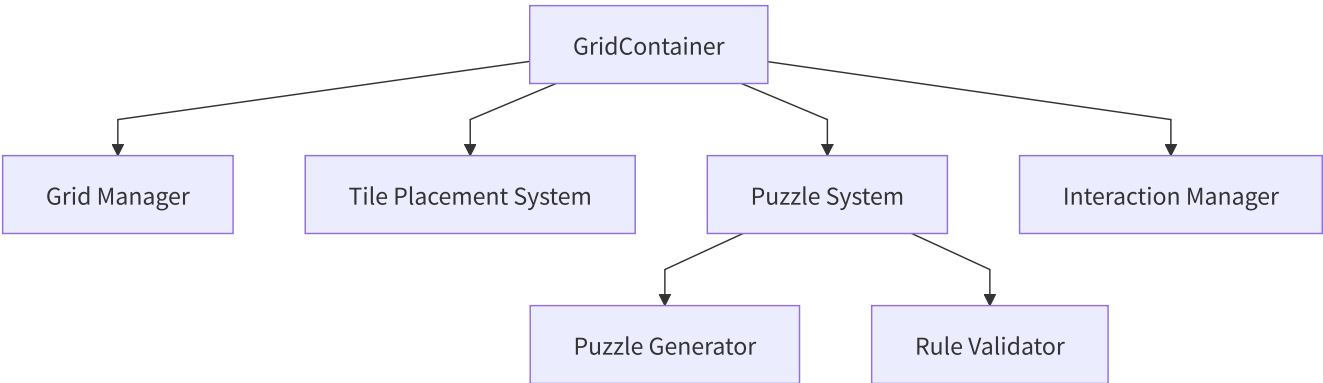


Figure 2. The core system architecture

As the panel presented as a game puzzle, this component serves as a container group capable of holding interactive tiles, and it supports both two-dimensional adjacency rules for 2D gameplay and three-dimensional adjacency rules for spatial gameplay.

Its core part is the **Grid Manager**, which is responsible for managing the component’s layout, rules, and puzzle pre-generation.

The **Tile Placement System** handles the placement and removal of tiles. The **Puzzle Generator** is responsible for initializing the puzzle by generating a random connected path as the solution, along with a series of transformation steps (see below). The **Rule Validator** checks whether the puzzle is completed—that is, whether a fully connected path between the start point and the end point has been formed. The **Interaction Manager** handles VR interactions.

5.2 Interactable Tile

The basic operational unit of the game is the individual **Tile**, which contains both a truth **value** and **logic**.

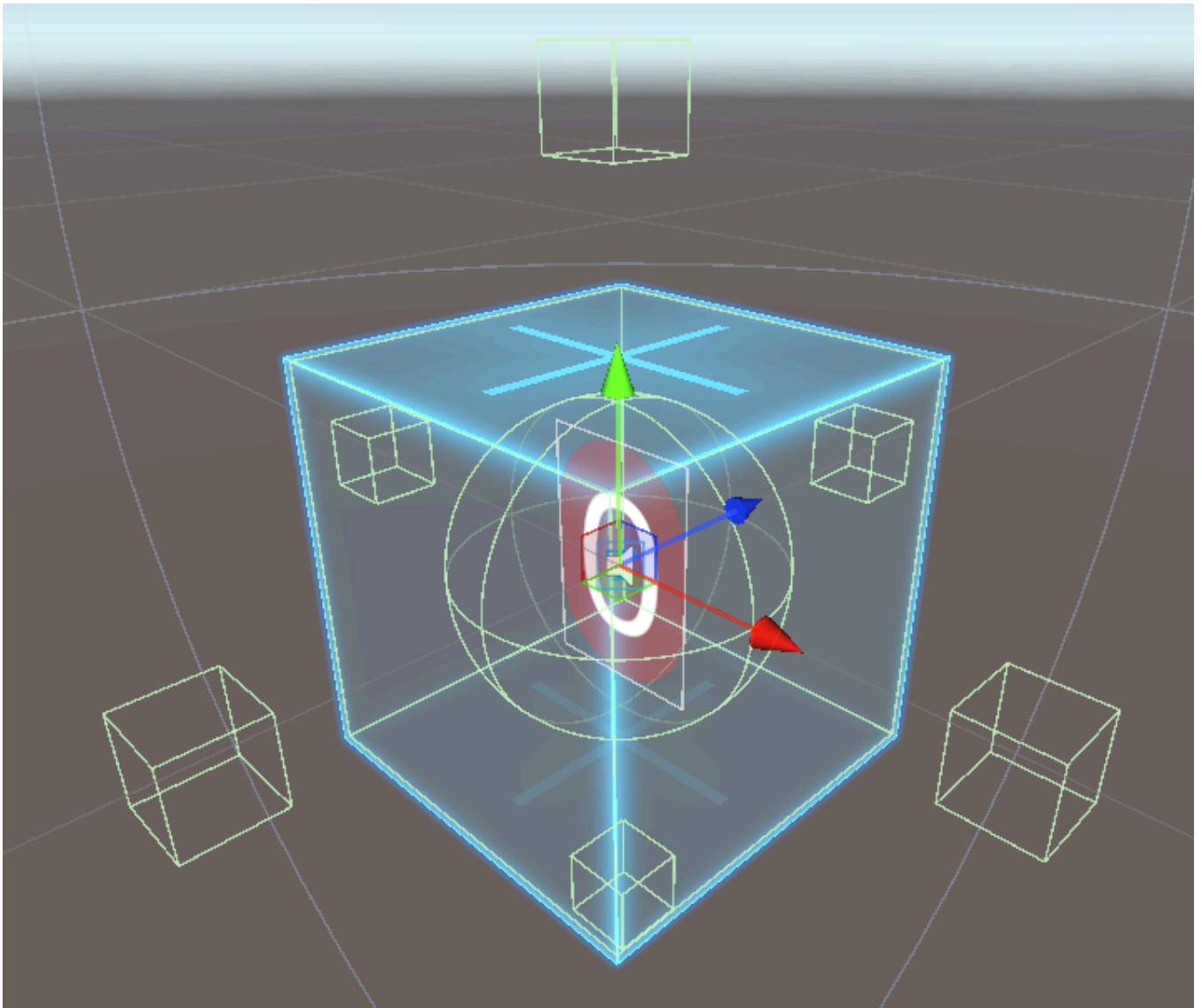


Figure 3. Tile with 0 value in Unity Editor

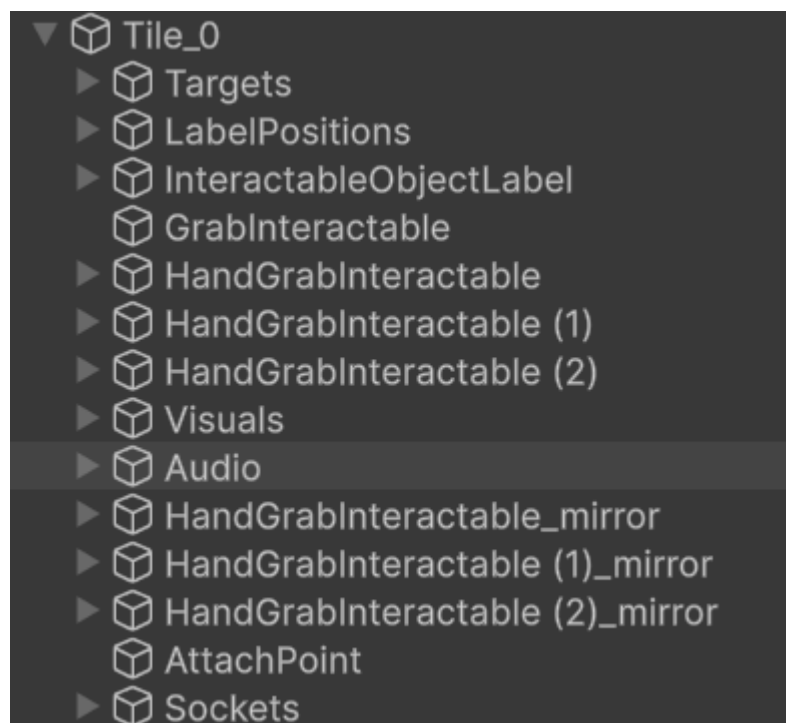


Figure 4. The structure of tile in Unity Editor

Inside each Tile, there is a **Label** (InteractableObjectLabel) used to display the currently stored value. When the truth value is **1**, the Label shows a green background; when the truth value is **0**, it shows a red background. The Label also supports adaptive sizing and orientation:

- it gradually scales down when the user approaches,
- gradually scales up when the user moves away,
- and caps at a predefined maximum size.

In addition, the Label rotates to always face the camera.

During interaction, the Label is temporarily hidden; after interaction stops for a certain period and the tile is being looked at, the Label reappears.

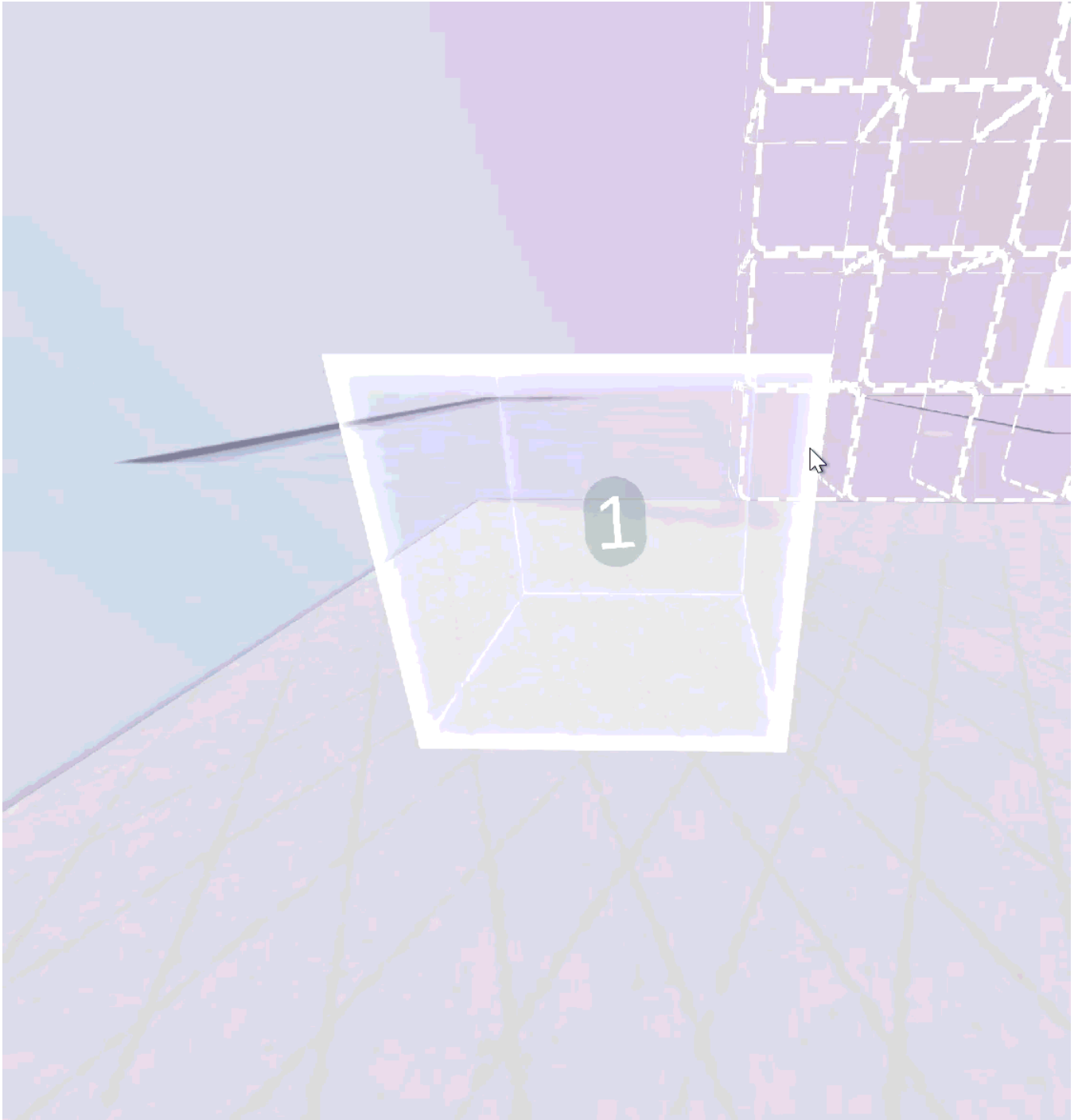
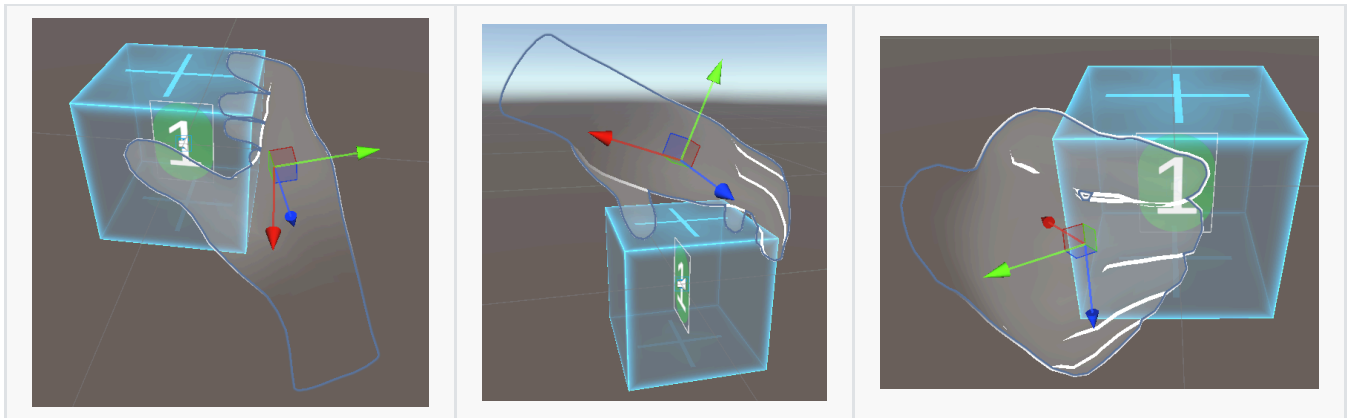


Figure 5. The label of tile

The Tile also provides multimodal feedback:

- When grasped using the controllers, its edges emit a slight glow, the controller delivers a light haptic vibration, and a pickup sound effect is played.
- When grasped using hand-tracking gestures, six different visual feedback patterns are shown depending on the specific grasp gesture and whether the left or right hand is used.
- When the Tile is released, a placement sound effect is played.

Table2. Three different grasp poses (right hand)



Tiles can connect with other Tiles of the same type to form various shapes. Value tiles can connect with other value tiles, and logic tiles can connect with other logic tiles.

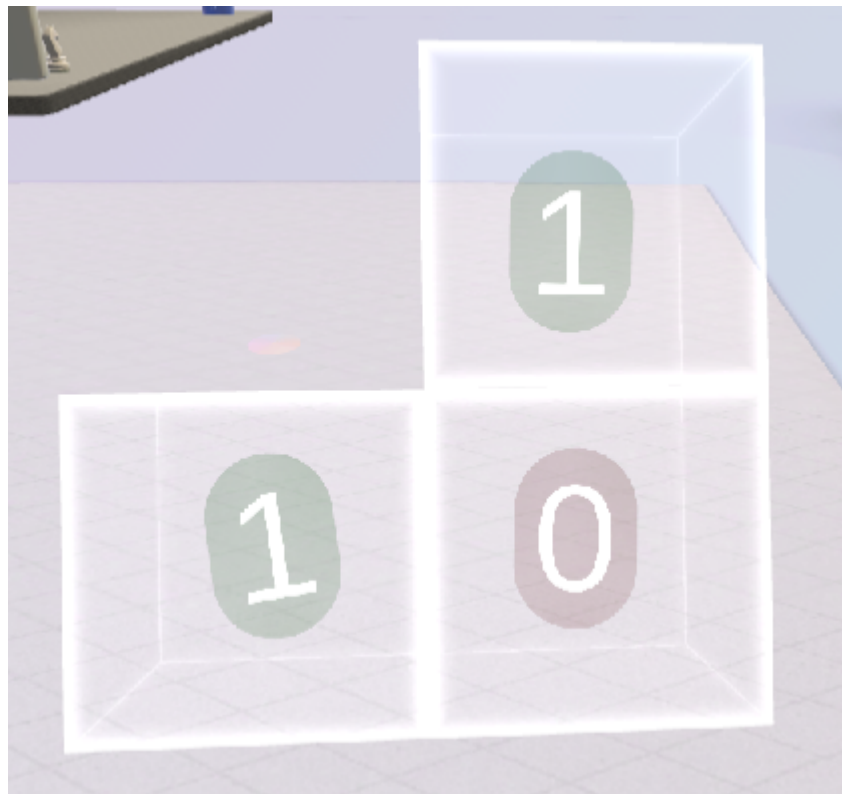


Figure 6. The combination of tiles

As shown in the figure 6 above, two Tiles with a truth value of 1 and one Tile with a truth value of 0 can be combined into a single unit. In VR mode, users can interact with the Tiles using either hand-tracking gestures or controllers to perform grasp actions.

When each hand grasps one Tile (or a Tile group) and the two are brought close to each other within a certain distance, the system checks whether a valid connection can be formed; if so, the connection is established once the user releases both hands.

Conversely, when both hands hold the two ends of a combined structure and move them apart (or twist them), exceeding a specified distance (or angle), the connection is broken.

The Tile connection system uses a non-traditional detection mechanism, different from simple collider-based detection or node-binding approaches. As shown in the figure, each Tile is equipped with a small collider on each of its six faces, each paired with a Socket Interactor, while the Tile itself has a Grab Interactable attached. This design effectively makes each Tile a six-sided plug, while each face also contains an invisible slot, ensuring that Tiles can be combined in arbitrary orientations and still snap precisely into the correct positions. Traditional collision-based detection would cause unpredictable physical interactions when multiple Tiles get close to one another, making proper alignment much more difficult to guarantee. The dimensions of the central Sphere Collider and the six external Box Colliders are carefully calculated to ensure stable alignment between sockets under any multi-Tile configuration or orientation, preventing incorrect snapping.

When a value Tile and a binary logic Tile are brought close together and compressed, they merge into a single unit, forming a logic Tile that carries a truth value. This combined logic Tile will revert into a value Tile when it receives another incoming value Tile.

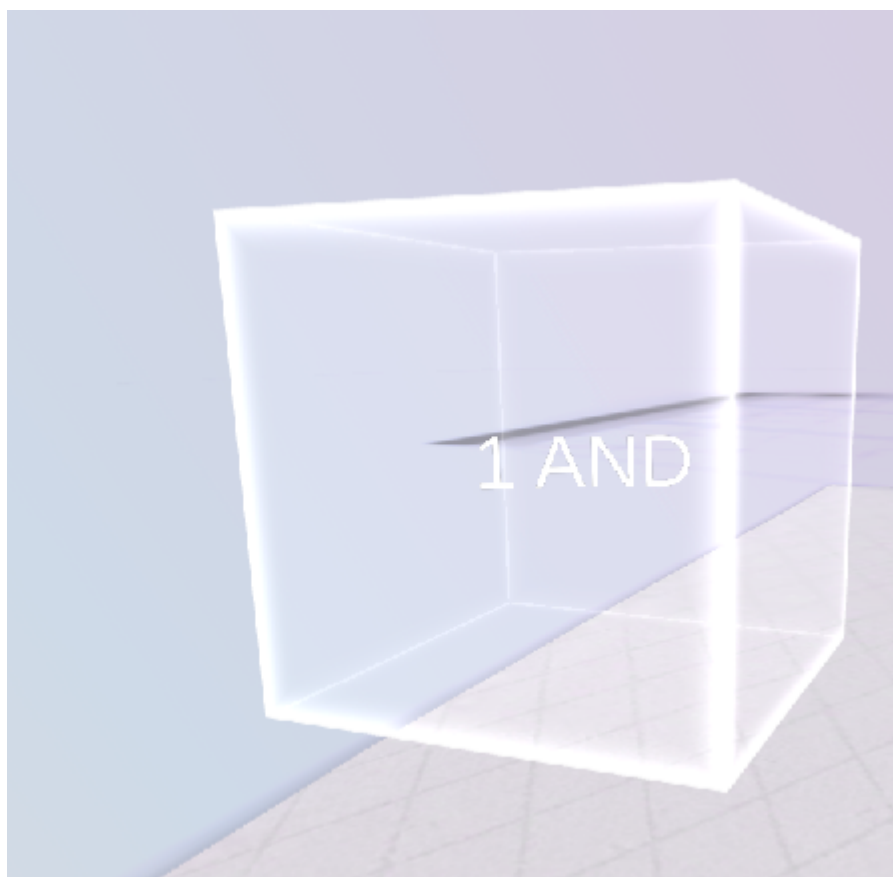


Figure 7. The combination of logic tile and truth tile

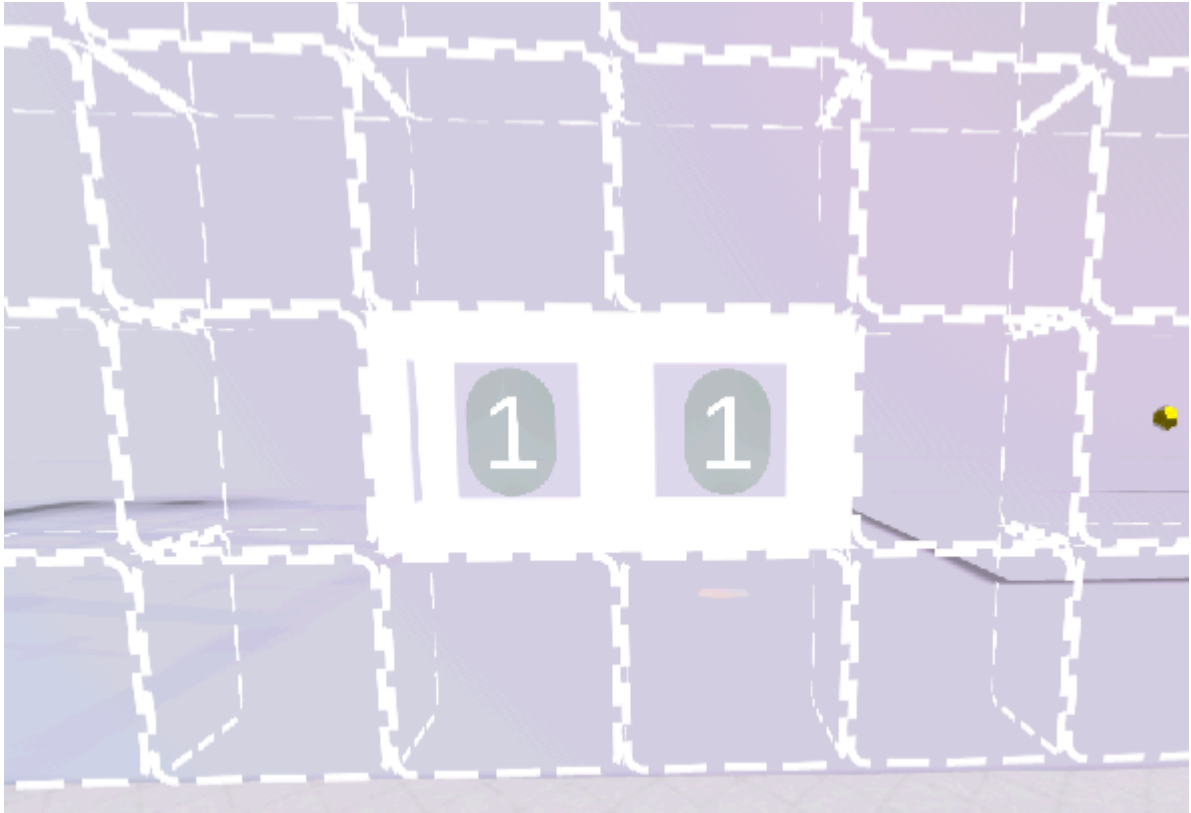


Figure 8. Tiles in grid container

As shown in the figure 8, this is the state in which Tiles are inserted into the Grid Container. In practice, when the game starts, the Grid Container automatically generates a Puzzle and fills it with different value Tiles. This Puzzle contains a visible start point and end point (randomly generated). (In the figure 9, the green Tile is the start, and the red Tile is the end; the screenshot is taken in Editor mode.) The program first generates the positions of the start and end points, ensuring that their coordinates fall within the index range of the container. It then generates a connected path between the start and the end—a path consisting entirely of Tiles with value 1 under the corresponding adjacency rules—ensuring that the Puzzle has at least one valid solution.

Next, along this path, the system randomly selects Tiles (including compound shapes formed by adjacent Tiles) and applies a sequence of random logic operations (such as NOT, XOR 1, AND 0), recording these operations. After a limited number of random logic transformations, the puzzle generation process is completed.

Finally, the system spawns the recorded logic operations and their corresponding value Tiles (or combined shapes) at randomly available positions in the scene. Depending on the difficulty settings, some of these combined Tile structures may not be allowed to be separated.

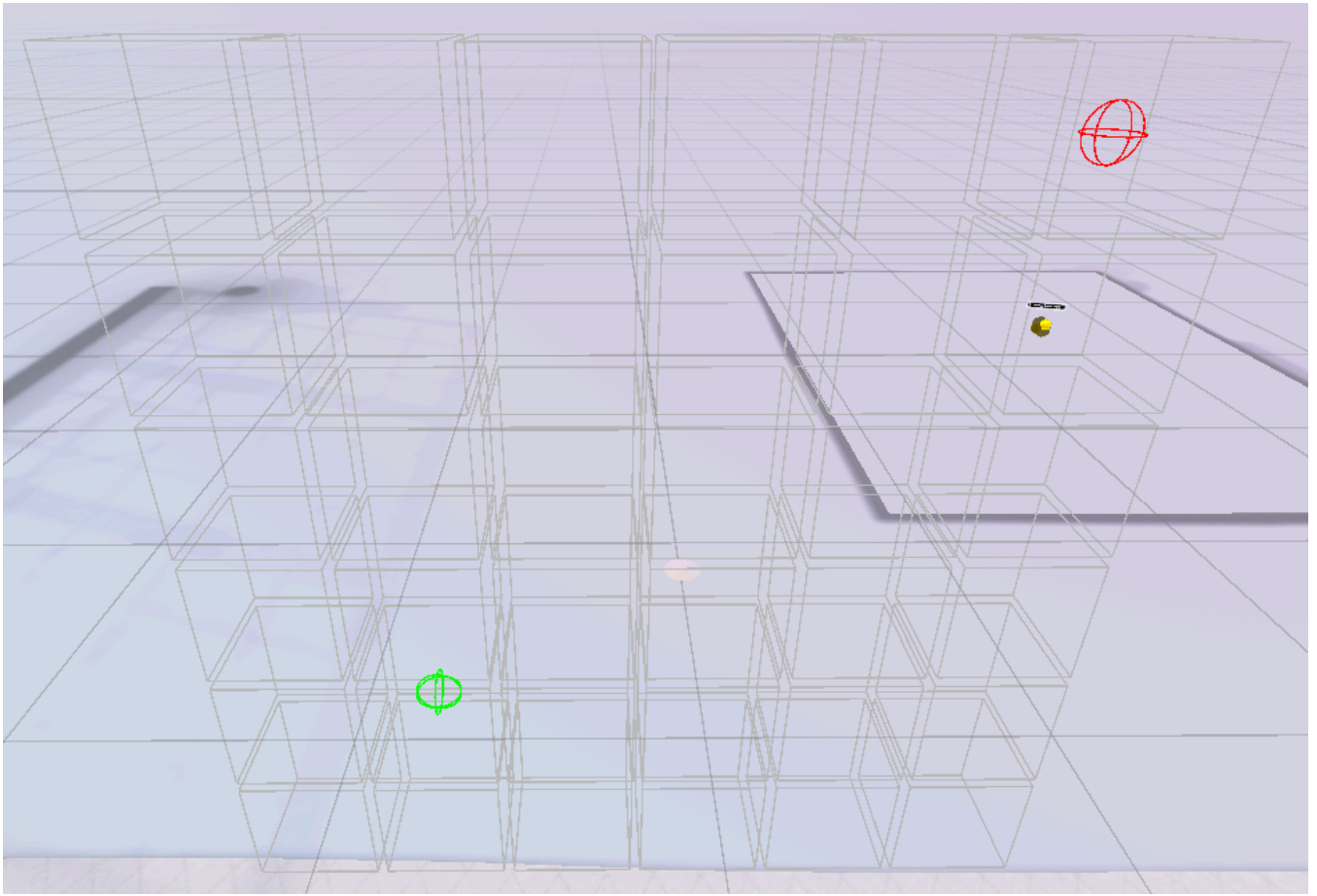


Figure 9. The start point and the end point of puzzle

During gameplay, the player is required to use the limited set of logic tiles and value tiles (or tile combinations) available in the scene to reverse this process and reconstruct the original puzzle-generation path (or any other valid solution). The **Rule Validator** uses a **BFS algorithm** to check whether there exists a valid connected path inside the Grid Container. If a valid path is detected, the game is won. Else if all tiles have been used and no valid path is found, the game is automatically considered failed and restarts. The system can be configured either to randomly generate a new puzzle or to restore the previously generated puzzle upon restart.

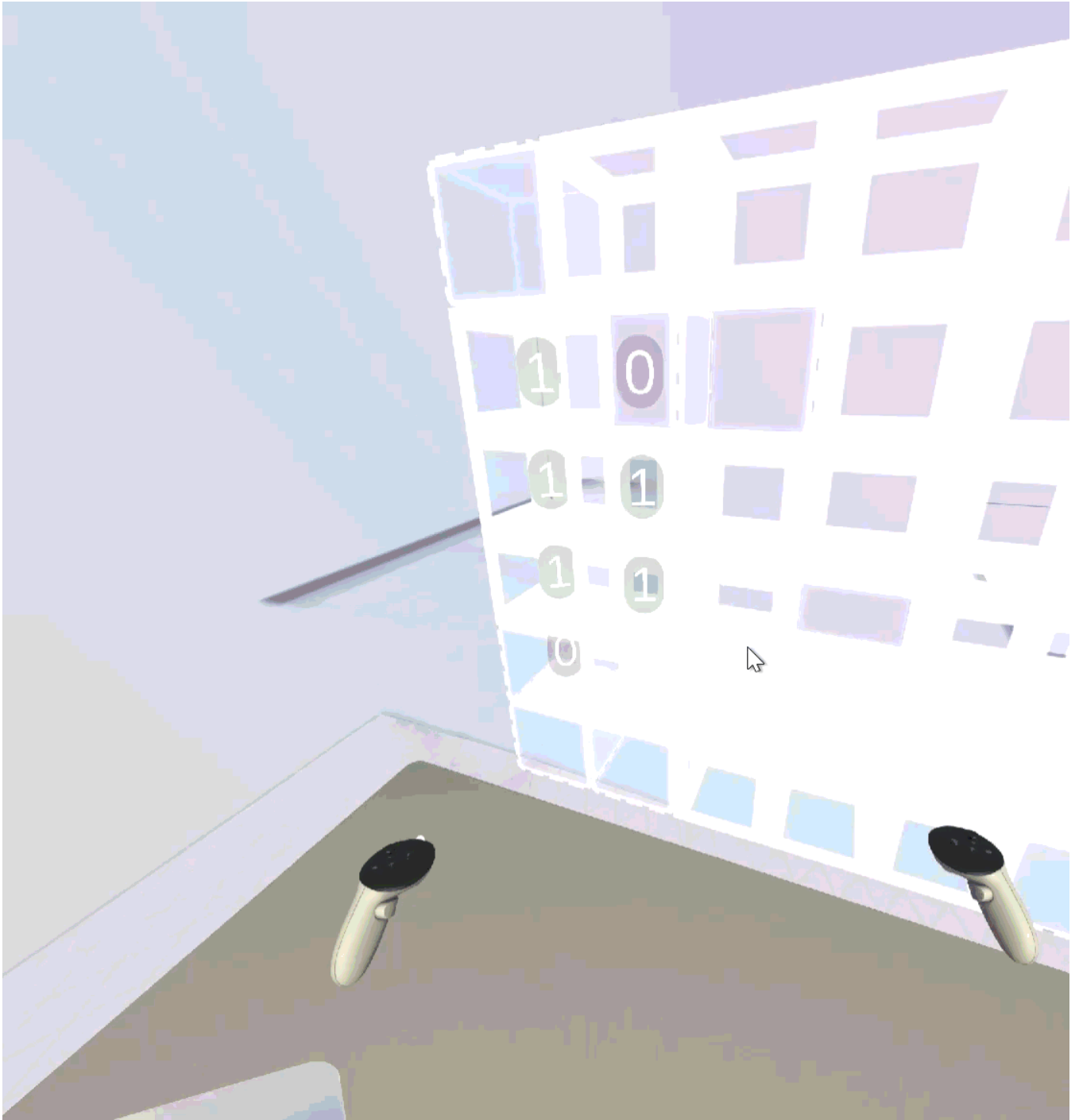


Figure 9. Fullfilled puzzle

5.3 Replay Value and Difficulty Curve

Since the number of possible puzzles is extremely large, we provide only a simplified estimation here. We consider a typical and most commonly studied setting: starting from the upper-left corner, one may move only to the right or downward until reaching the lower-right corner (the classical *lattice path* problem, also a fundamental result in BFS and combinatorics).

In this setting, the total number of distinct paths from $(0, 0)$ to (m, n) is:

$$\binom{m+n}{m} = \binom{m+n}{n}$$

That is, the path consists of m downward steps and n rightward steps, for a total of $m + n$ moves. Any permutation of these downward and rightward steps corresponds to a unique path. It is not difficult to see that this binomial coefficient grows very rapidly as $m + n$ increases—approaching exponential growth.

When m and n increase simultaneously (e.g., a square grid):

Number of paths $\approx \frac{4^k}{\sqrt{\pi k}}$ where $m = n = k$

which follows from a Stirling-based approximation. When m or n is significantly larger than the other, the number of paths grows even more closely to an exponential rate.

Table3. relation between grid scale and number of paths

Grid (m × n)	Number of paths $\binom{m+n}{m}$	规模级别
2×2	6	1
3×3	20	10
4×4	70	10
5×5	252	10 ²
10×10	184, 756	10 ⁵

However, in practice the puzzle has far more possible path configurations, due to the randomized start and end points as well as the allowance of branching paths. Therefore, even for a 5 × 5 Container, the number of possible paths is already on the order of 10³. When the randomized logic operations are further taken into account, it becomes clear that encountering the same puzzle twice is virtually impossible.

On the other hand, increasing the number or length of possible paths does not significantly affect the difficulty curve of the game. Intuitively, no matter how long or complex the underlying path is, if only single-bit negation operations are applied on top of it, the player can always solve the puzzle almost instantly. It is therefore evident that the primary factor influencing the difficulty curve is the number of logic operations applied during puzzle generation—especially region-based operations and the allowance of repeated operations on the same position. The number of logic operations increases the puzzle difficulty approximately in a quadratic manner.

6.Conclusion and Future Work

Over the past ten weeks of development, the core prototype of Binary Path has been largely completed. This includes the implementation of a grid-management framework, an interactive Tile system, a full puzzle-generation and validation mechanism, and multimodal interaction within a VR environment. The development process did not fully follow the initial plan, and many new challenges were discovered along the way (such as collision handling when multiple Tiles approach each other), along with several novel solutions (such as the socket-based connection system). The next stage of work will shift toward more concrete development tasks: scene construction, 3D asset creation, level design, save/load functionality, and the multiplayer mode.

In terms of game mechanics, an innovative idea will guide future work: **using QR codes to store predefined puzzles**. I hope to present the game's levels in the form of a *Player Handbook* and explore the feasibility of this model in VR-based entertainment and education. If possible, I aim to extend this concept into an MR-based experience—for example, generating Tiles on the real-world floor, embedding the Container into physical walls, and loading puzzles simply by tapping on printed pages. Such an approach would undoubtedly enrich the user experience and enhance interaction among different users.



Figure 10. QR code detection

7.Schedule

Table 4. Timeline of POC implement

Week	Tasks	Description
1	-	
2	-	
3	Proposal	
4	Proposal & Base Frame & Hub Prototype	1. Establish the project baseline: URP, XR Interaction Toolkit, input mapping (grab/rotate/pointer). 2. Hub scene v0: placeholder for 2-3 portals, scene loading (Additive), a "home portal" for returning to the Hub. 3. Performance budget and template: target 90Hz (72Hz if necessary), fixed render resolution, baked lighting, platform settings (Quest 3). 4. Version control and continuous integration: Git (branching conventions)
5	Grid System	Grid System/Tile/Bit System • Implement grid system (0/1 state). • Add tile placement (snap-to-grid). • Implement path checking (DFS/BFS).
6	Limits	• Add Obstacle Tiles (non-placeable). • Introduce limited tile inventory. • Add error feedback (red highlight, vibration).

7	Logic tiles	• Add Logic tile (toggle 0 ↔ 1). • Add usage limits per level. • Add optional timer for scoring.
8	Hub Polish + Achievements + Light Narrative	• Hub update with Quest Scroll showing objectives. • Add basic SFX/particle feedback for wins. • Add local achievement system (JSON).
9	UX Test + Interaction/Performance Improvements	• Conduct small usability test (1-3 users). • Improve grab/snap feel (thresholds, smoothing). • Fix common issues (misplacement, slow TEST cycle). • Optimize performance (batching, LOD, baking).
10	POC Finalization & Demo Preparation	• Build playable POC APK. • Record video. • Write POC report
11	-	
12	-	

Table 5. Timeline of POC implement

Week	Tasks	Description
1	Multiplayer Core Setup	• Implement Host/Client flow and synchronized scene loading. • Create a simple multiplayer lobby (Host Game / Join Game). • Ensure both players spawn correctly in the same VR space.
2	Tile Placement Synchronization	• Synchronize tile placement across clients (grid updates must be network-authoritative). • Implement turn-based system • Add rejection rules for illegal moves
3	Win Conditions & Game Rules	• Implement win-condition logic • Add connected-path checking • Add optional rule constraints
4	VR User Experience & Gameplay Feel	• Improve clarity of turn-switch feedback • Add multimodal feedback
5	Match Flow, Reset, & Session Stability	• Implement full board reset + automatic first-player swap. • Handle network errors
6	Final Polish & Multiplayer MVP Build	• Multi-device testing for reliability • Export final Multiplayer MVP build.

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